

Quiz (Carolina Reaper)

Q1. Solve $x^2 + 5x + 6 = 0$ by factoring.

- (A) $(x + 2)(x + 3) = 0$
- (B) $(x - 2)(x - 3) = 0$
- (C) $(x + 2)(x - 3) = 0$
- (D) $(x - 2)(x + 3) = 0$
- (E) $(x + 1)(x + 4) = 0$

Q2. Solve $x^2 - 7x + 12 = 0$ by factoring.

- (A) $(x - 3)(x - 4) = 0$
- (B) $(x + 3)(x + 4) = 0$
- (C) $(x - 3)(x + 4) = 0$
- (D) $(x + 3)(x - 4) = 0$
- (E) $(x - 2)(x - 5) = 0$

Q3. Solve $x^2 + 2x - 15 = 0$ by factoring.

- (A) $(x + 5)(x - 3) = 0$
- (B) $(x - 5)(x + 3) = 0$
- (C) $(x + 5)(x + 3) = 0$
- (D) $(x - 5)(x - 3) = 0$
- (E) $(x + 2)(x - 6) = 0$

Q4. Solve $x^2 - 4x - 21 = 0$ by factoring.

- (A) $(x - 7)(x + 3) = 0$
- (B) $(x + 7)(x - 3) = 0$
- (C) $(x + 7)(x + 3) = 0$
- (D) $(x - 7)(x - 3) = 0$
- (E) $(x + 2)(x - 8) = 0$

Q5. Verify the solution to $x^2 + 2x - 6 = 0$ is $x = -1 \pm \sqrt{7}$.

- (A) The solution is correct.
- (B) The solution is incorrect.
- (C) More information is needed.
- (D) The equation has no solution.
- (E) The equation has infinitely many solutions.

Q6. Solve $x^2 + 10x + 24 = 0$ by factoring.

- (A) $(x + 4)(x + 6) = 0$
- (B) $(x - 4)(x - 6) = 0$
- (C) $(x + 4)(x - 6) = 0$
- (D) $(x - 4)(x + 6) = 0$
- (E) $(x + 2)(x + 8) = 0$

Q7. Solve $x^2 - 9x + 20 = 0$ by factoring.

- (A) $(x - 4)(x - 5) = 0$
- (B) $(x + 4)(x + 5) = 0$
- (C) $(x + 4)(x - 5) = 0$
- (D) $(x - 4)(x + 5) = 0$
- (E) $(x - 2)(x - 7) = 0$

Q8. Solve $x^2 + x - 12 = 0$ by factoring.

- (A) $(x + 4)(x - 3) = 0$
- (B) $(x - 4)(x + 3) = 0$
- (C) $(x + 4)(x + 3) = 0$
- (D) $(x - 4)(x - 3) = 0$
- (E) $(x + 2)(x - 5) = 0$

Open Ended Questions (FRQ)**Q9.** Solve the equation $x^2 + 5x + 6 = 0$ by factoring and verify the solutions.**Q10.** Solve the equation $x^2 - 7x - 18 = 0$ by factoring and verify the solutions.**Chef's Tips**

- Make sure to check the work for the free response questions.
- Verify that the students have factored the equations correctly.
- Remind the students to simplify their answers.